

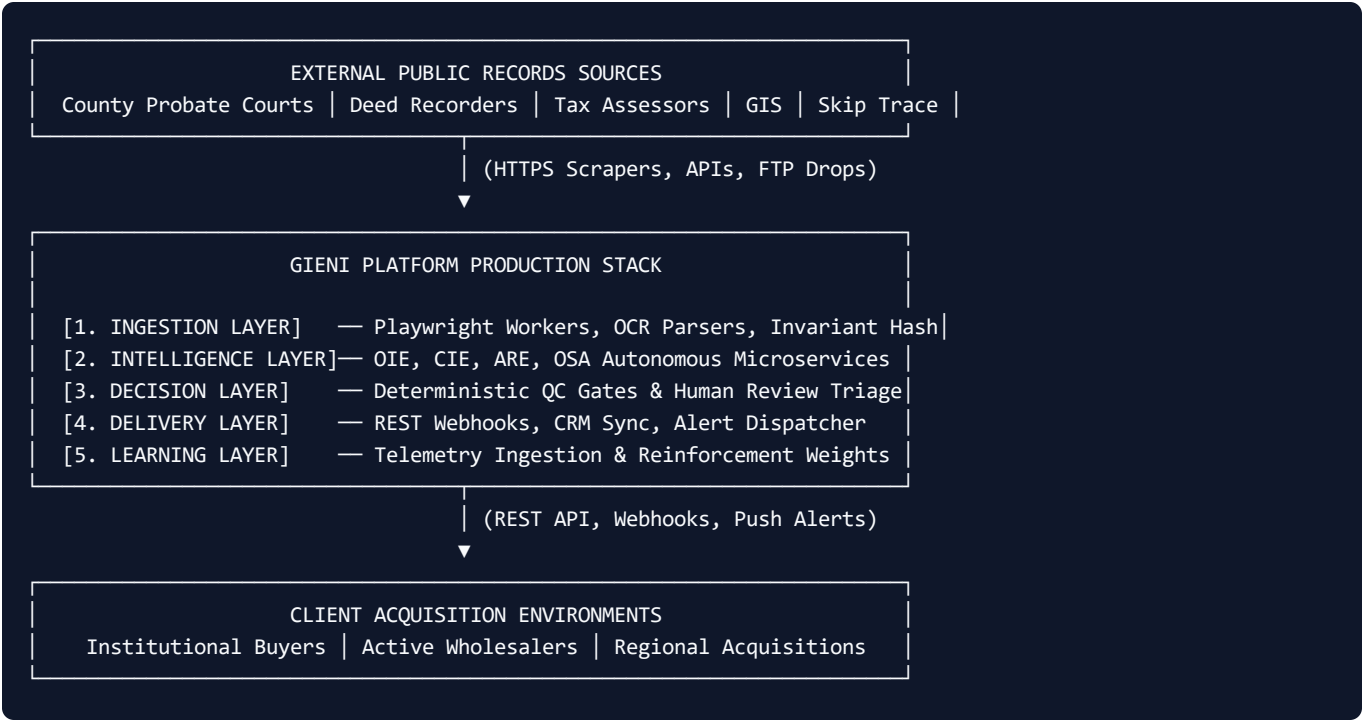
Giени Platform Reference Architecture

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Production System Design for the Acquisition Decision Intelligence Platform
Classification: Canonical Engineering Architecture | Version: 2.0 | Owner: CTO / Systems
Architecture | Status: Production Approved



1. Platform Overview & System Context

Canonical technical blueprint for the Giени Acquisition Decision Intelligence Platform. This document defines the production microservice boundaries, event schemas, deployment topology, lifecycle state machine, failure recovery runbooks, and security governance required for zero-defect execution across hundreds of county jurisdictions.



2. Service Ownership & Deployment Architecture

The platform is implemented as nine decoupled, independently deployable microservices backed by an asynchronous event bus and isolated persistence stores:



1. Property Service

- Purpose: Reconciles candidate addresses from court petitions against county assessor rolls & GIS layers.
- Primary Database: properties, property_assessments (PostgreSQL + PostGIS)
- Events Published: Property.Identified
- Events Consumed: ProbateCase.Created
- Dependencies: County GIS shapefile endpoints, Assessor API
- Latency SLA: < 120s per case docket.



2. Ownership Service

- Purpose: Analyzes recorded deed instruments to extract vesting, verify grantor/grantee chains, and compute Net Equity.
- Primary Database: ownership_events, title_chains, encumbrances (PostgreSQL)
- Events Published: Ownership.Updated
- Events Consumed: Property.Identified
- Dependencies: County Recorder OCR engine, Title plant search API
- Latency SLA: < 180s per property.



3. Control Service

- Purpose: Maps stakeholder relationships, models multi-heir social networks, and identifies the de-facto decision-maker.
- Primary Database: stakeholders, control_assessments, graph_edges (PostgreSQL + Neo4j)
- Events Published: Control.Changed
- Events Consumed: Ownership.Updated
- Dependencies: Heirs schedule NLP extractor, Skip trace utility API
- Latency SLA: < 90s per estate.



4. Authority Service

- Purpose: Tracks court docket minutes, verifies statutory Letters Testamentary/Administration, and classifies signatory capacity.
- Primary Database: authority_candidates, court_dockets, authority_assessments (PostgreSQL)
- Events Published: Authority.Updated
- Events Consumed: Control.Changed
- Dependencies: Probate court docket scrapers
- Latency SLA: < 60s per docket update.



5. Opportunity Service

- Purpose: Synthesizes net equity against Deal Friction Score (DFS) to compute Viability Score (0–100) and Priority Tier.
- Primary Database: opportunities, opportunity_assessments (PostgreSQL)
- Events Published: Opportunity.Scored
- Events Consumed: Authority.Updated, Ownership.Updated
- Dependencies: LightGBM Valuation microservice
- Latency SLA: < 15s per scoring pass.



6. Evidence Service

- Purpose: Manages chain-of-custody artifacts, PDF document storage, and SHA-256 hash validation.
- Primary Database: evidence_sources, claim_evidence (PostgreSQL + S3 Blob Store)
- Events Published: Evidence.Ingested
- Events Consumed: ProbateCase.Created, Ownership.Updated
- Dependencies: AWS S3 / Cloudflare R2
- Latency SLA: < 500ms per document hash commit.



7. Client Service

- Purpose: Governs single-partner county exclusivity rules, capacity caps, buy-box matching, and billing states.
- Primary Database: clients, territory_agreements (PostgreSQL)
- Events Published: Client.ConfigUpdated
- Events Consumed: None (Root Administration)
- Dependencies: Stripe API
- Latency SLA: < 200ms.



8. Delivery Service

- Purpose: Dispatches certified opportunity packages to partner CRMs via webhooks and fires Twilio SMS Priority A alerts.
- Primary Database: opportunity_deliveries, webhook_logs (PostgreSQL)
- Events Published: Opportunity.Delivered
- Events Consumed: Opportunity.QCCertified
- Dependencies: Twilio API, SendGrid, Partner CRM Webhook Endpoints
- Latency SLA: < 5s from QC approval to dispatch.



9. Workflow & Orchestration Service

- Purpose: Coordinates the 14-stage DAG lifecycle state machine, timeout timers, and human exception routing.
- Primary Engine: Temporal Workflow Engine + Redis BullMQ
- Events Published: Pipeline.ExceptionRouted
- Events Consumed: All lifecycle events
- Dependencies: PostgreSQL, Redis
- Latency SLA: Sub-second state transitions.

3. Opportunity Lifecycle Progression Matrix

Part 1: Research & Intelligence Stages (Discovered → QC Certified)

STATE	CREATED BY	REQUIRED DATA CONTRACT	NEXT PERMITTED STATES
DISCOVERED	Ingestion Worker	Valid Case #, Filing Date, Decedent Name	PROPERTY_IDENTIFIED, EXCEPTION
PROPERTY_IDENTIFIED	PRA Agent	Reconciled APN, Situs Address, PAS ≥ 70	OWNERSHIP_RESOLVED, EXCEPTION
OWNERSHIP_RESOLVED	OIA Agent	Deed Vesting, Encumbrance Balances, Equity	CONTROL_MAPPED, DISQUALIFIED
CONTROL_MAPPED	CIA Agent	Heir Tree, De-Facto Controller, Archetype	AUTHORITY_RESOLVED, EXCEPTION
AUTHORITY_RESOLVED	ARA Agent	Letters Status, Authority Tier (1-4), Power	SCORED, EXCEPTION
SCORED	OSA Agent	Composite Score (0-100), DFS, Priority	QC_CERTIFIED, DISQUALIFIED
QC_CERTIFIED	QC Agent / QA Lead	Passed 6 Verification Gates, SHA Hashes	DELIVERED, EXCEPTION

Part 2: Commercial & Disposition Stages (Delivered → Archived)

STATE	CREATED BY	REQUIRED DATA CONTRACT	NEXT PERMITTED STATES
DELIVERED	Delivery Service	CRM Webhook ACK / Batch Staging Commit	CONTACTED, EXCEPTION
CONTACTED	Partner Specialist	First-Touch Timestamp, Channel (Call/SMS)	APPOINTMENT, REJECTED
APPOINTMENT	Partner Specialist	Scheduled Date, Time, Decision-Maker Conf	OFFER_PRESENTED, REJECTED
OFFER_PRESENTED	Partner Specialist	Written Cash Offer Amount, Terms Logged	CONTRACT_EXECUTED, REJECTED
CONTRACT_EXECUTED	Closing Attorney	Signed Purchase Agreement, Escrow Opened	CLOSED_WON, CLOSED_LOST
CLOSED_WON	Closing Attorney	Title Recorded, Wholesale Fee Realized	ARCHIVED
CLOSED_LOST	Closing Attorney	Cancellation Reason Code Logged	ARCHIVED
ARCHIVED	Feedback Agent	Telemetry Committed to ML Training Set	(Terminal State)

4. Canonical Event Contracts (JSON Payloads)

Every inter-service event emitted over the Apache Kafka / EventBridge message bus must strictly conform to these validated JSON contracts:

```
// 1. Property.Identified Event Contract
{
  "eventType": "Property.Identified",
  "eventId": "e9b21f3a-6410-4e31-bc8d-0c84f3918a22",
```

```

"timestamp": "2026-09-18T10:14:22Z",
"caseId": "c41890e1-7711-4fa2-9011-fec81902ba11",
"propertyId": "p771891a-1122-48a0-99c0-bb8492019910",
"countyFips": "48201",
"apn": "041-280-001-002",
"situsAddress": {
  "street": "1428 Elm Street",
  "city": "Houston",
  "state": "TX",
  "zip": "77002"
},
"pasScore": 94.5,
"attributionMethod": "GIS_NAME_STREET_MATCH"
}

```

```

// 2. Ownership.Updated Event Contract
{
  "eventType": "Ownership.Updated",
  "eventId": "f81829aa-2091-4c11-9a99-de8192011928",
  "timestamp": "2026-09-18T10:16:05Z",
  "propertyId": "p771891a-1122-48a0-99c0-bb8492019910",
  "vestingType": "SOLE_FEE_SIMPLE",
  "deceasedTitleholder": "Robert E. Johnson",
  "avmMarketValue": 345000.00,
  "totalEncumbrances": 85000.00,
  "netEquity": 260000.00,
  "equityPct": 0.7536,
  "ownershipComplexityScore": 25
}

```

```

// 3. Authority.Updated Event Contract
{
  "eventType": "Authority.Updated",
  "eventId": "a71829bb-1188-4e00-88aa-109283746192",
  "timestamp": "2026-09-18T10:18:12Z",
  "caseId": "c41890e1-7711-4fa2-9011-fec81902ba11",
  "authorityTier": "TIER_1_CONFIRMED",
  "lettersStatus": "ISSUED",
  "powerScope": "FULL_INDEPENDENT_ADMINISTRATION",
  "fiduciary": {
    "personId": "per_991823-aa11",
    "name": "Sarah Johnson Miller",
    "relationship": "Daughter",
    "primaryPhone": "+18325551928",
    "phoneCarrierValid": true,
    "dncScrubbed": true
  },
  "attorneyQuarantined": true
}

```

```

// 4. Opportunity.Scored Event Contract
{
  "eventType": "Opportunity.Scored",

```

```

"eventId": "b91827cc-3321-4999-bb11-491827364510",
"timestamp": "2026-09-18T10:19:44Z",
"opportunityId": "opp_441890-ee88",
"compositeViabilityScore": 88,
"priorityTier": "PRIORITY_A",
"dealFrictionScore": 18,
"dispatchSLA": "FLASH_ALERT_4_HOUR"
}

```

5. Failure Recovery & Dead-Letter Queue (DLQ) Architecture



Deterministic Exception Recovery Runbook

1. Municipal Scraper Failure: Exponential backoff (retry after 1m, 5m, 15m) with automatic residential proxy rotation. If 3 retries fail, alert PagerDuty on-call data engineer.
2. OCR Document Parsing Failure: Unreadable or corrupted PDF dockets are diverted to the Tasks & Exceptions database tagged with "OCR Corruption" for manual transcription.
3. Title & Deed Conflict: If recorded deeds show unprobated intermediate grantors (broken deed chain), job is isolated and tagged "Title Exception".
4. Authority Ambiguity: Cases with competing heir petitions or caveats are tagged "Authority Unresolved" and routed to legal review queue.
5. Partner Webhook Failure: 5 retries with exponential backoff over 2 hours. If endpoint fails persistently, fallback alert is dispatched to partner via SMS and email, and payload is routed to Dead-Letter Queue (DLQ).

6. Formalized Graph Topology (Neo4j / Amazon Neptune)

Canonical Nodes & Edge Semantics

- Nodes: Property, Person, Trust, Estate, Authority Candidate, Client.
- Directed Edges:
 - (:Person)-[:OWNS]->(:Property)
 - (:Person)-[:CONTROLS]->(:Property)
 - (:Person)-[:TRUSTEE_OF]->(:Trust)
 - (:Person)-[:BENEFICIARY_OF]->(:Estate)
 - (:Person)-[:REPRESENTS]->(:Estate) [Legal Counsel]
 - (:AuthorityCandidate)-[:AUTHORITY_OVER]->(:Property)
 - (:Opportunity)-[:DELIVERED_TO]->(:Client)
- Strategic Impact: Graph traversal algorithms uncover unlisted heirs, unrecorded trust instruments, and hidden authority pathways across multi-jurisdiction estates.

7. Security, RBAC & Data Governance

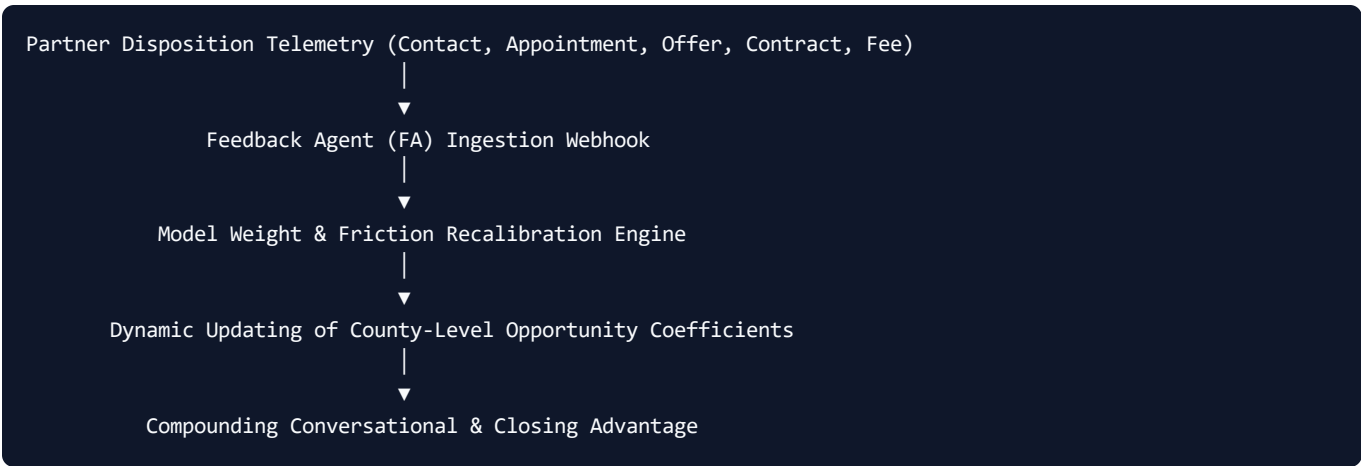


Four-Tier Data Classification Standard

1. Public Data: Court docket numbers, recorded deed instruments, tax rolls. Storage: Standard PostgreSQL / S3.
2. Enriched Data: CASS-standardized addresses, skip-traced phone lines, DNC check metadata. Storage: Encrypted at rest (AES-256).
3. Internal Intelligence: OIE/CIE/ARE assessments, Viability Scores, Deal Friction calculations. Storage: Strictly isolated proprietary database.
4. Client Confidential: Partner disposition notes, wholesale fees realized, offer amounts. Storage: Partitioned per tenant with Row-Level Security (RLS).

Role-Based Access Control (RBAC): Research Specialist (read/write dockets and deeds), Operations Lead (full access to QC and exception queues), Admin / CTO (infrastructure and model parameters), and Client Partner (read-only access restricted to licensed territory).

8. The Outcome Telemetry & Continuous Learning Loop



The Gieni Platform Reference Architecture provides the canonical technical contract uniting autonomous AI agents, deterministic municipal logic, and institutional client outcomes.